

Four questions this dissertation refused to outsource

left — the question; right — what the field's standard answer was structurally underspecified to recover for this problem

Is the field's autonomous stability criterion the right object to measure for a driven reservoir?

$\rho(W) < 1$ verifies a static property of the recurrent matrix; it does not characterize how the reservoir contracts under the input it actually receives. For a reservoir whose job IS the driven response, the autonomous criterion is structurally underspecified.

→ Experiment A

Can the reservoir's choice be justified from a theorem about the data, not from architectural fashion?

The Maass-Markram licensing argument is non-constructive for a specific dataset — it tells you reservoirs can separate, not whether mine does for these ERPs. Takens' theorem makes a measurable claim about the data the reservoir must accommodate; I wanted that measurement.

→ Experiment B

Can the operating hyperparameter be derived rather than tuned?

Validation-accuracy-driven selection optimizes for the metric, not for the underlying information-theoretic property. A measured memory-capacity peak provides a stable, theoretically-anchored operating point that does not move when the downstream task changes.

→ Experiment C

Do clinical effects survive the strictest possible null — channel permutation — at the per-disorder level?

Parametric p-values assume a model of the noise structure that EEG channel selection does not satisfy. A channel-permutation null assumes nothing about the channels; it tests the claim the author actually wants to make.

→ Experiment D

Each question is answered, with measurement, in the experiments that follow.

Five methodological refusals. Each names a default the field accepts and that this dissertation chose to refuse.

What I refused to do

Report trial-level cross-validation (typically inflates clinical-EEG accuracy by 10-15 percentage points by leaking same-subject correlation across train and test).

Use parametric p-values that assume noise-structure models EEG channel selection does not satisfy.

Select β by maximizing validation accuracy.

Treat autonomous $p(W) < 1$ as sufficient evidence that the reservoir is in the usable regime.

Justify the reservoir architecturally through Maass-Markram alone (a non-constructive existence theorem).

What I did instead, and why

Subject-level StratifiedGroupKFold across every classification claim in the dissertation. Trial-level numbers were never the headline.

→ Chapter 5; Exp D

Per-trial channel-permutation null (Exp D) that assumes nothing about channels and tests the spatial claim the dissertation actually makes.

→ Exp D

Select β within the measured memory-capacity plateau (Exp C), anchored to a biological time-constant argument ($1/\beta \approx 20$ steps matches the α -period). The choice does not move when the downstream task changes.

→ Exp C

Measure driven λ_1 along 3,165 real ERP trajectories (Exp A). The reservoir is in the usable regime BECAUSE we measured it there.

→ Exp A

Measure the data's Takens dimension and verify the reservoir's state space exceeds it by 16-64 $\times$ (Exp B). The justification is data-specific, not generic.

→ Exp B

Each refusal is a rigor commitment. Each commitment is in the experiments above.

From these thinkers I learned how to ask — not just what to cite.

Intellectual provenance, not citation dump. Each entry names a question I would not have asked without them.

Takens, Sauer, Yorke

embedding theory

Takens 1981 · Sauer-Yorke-Casdagli 1991

From them, I learned to ask...

...whether the data's latent dynamics could be reconstructed from a sufficiently rich observation function — and whether that reconstruction is measurable on real ERPs.

Exp B

Maass, Natschläger, Markram

liquid state machines

Maass, Natschläger, Markram 2002

From them, I learned to ask...

...what kind of dynamical state space a useful reservoir must possess, and what separation property it must exhibit for distinct inputs.

Exp B premise · Ch. 5 architecture

Jaeger, Lukoševičius

Echo State Network theory

Jaeger 2001 · Lukoševičius-Jaeger 2009

From them, I learned to ask...

...whether the reservoir's response to input was consistent enough to be readable — and how to measure that consistency rather than assume it.

Exp A

Oseledec, Benettin, Wolf

Lyapunov spectra & numerical methods

Oseledec 1968 · Benettin et al. 1980 · Wolf et al. 1985

From them, I learned to ask...

...what stability MEANS along a driven trajectory — and how to compute it when the analytic Jacobian is undefined at threshold events.

Exp A method

Bullmore, Sporns

graph-theoretic neuroscience

Bullmore & Sporns 2009

From them, I learned to ask...

...whether the brain's spatial structure could be operationalized as a measurable graph — and whether that graph carries clinical information that survives nullification.

Ch. 5 graph construction · Exp D

Marder, Krakauer et al.

philosophy of neuroscience

Marder 2015 · Krakauer, Ghazanfar, Gomez-Marin, MacIver, Poeppel 2017

From them, I learned to ask...

...whether benchmark accuracy is what we should optimize for in clinical neuroscience — and what we lose when interpretability becomes the trade variable.

Methodological Refusals · J contributions

Theorem scaffold: every architectural choice traces to a named theorem the dissertation measures

Theorems

Takens' embedding theorem

Takens (1981)
A delay reconstruction of
sufficient dimension preserves
the latent attractor's topology.

Maass-Markram separation

Maass, Natschläger, Markram (2002)
A high-dimensional recurrent
network maps distinct inputs
to distinct trajectories.

Oseledec multiplicative ergodic theorem

Oseledec (1968)
Along an ergodic trajectory,
the Lyapunov spectrum is
well-defined.

Echo State Property

Jaeger (2001)
For any input, the reservoir
converges to an input-driven
manifold.

Jaeger memory capacity

Jaeger (2001)
Total recall MC is bounded
above by N ; peaks at an
interior leak rate.

Architectural choices they license

**Use a reservoir, sized to exceed
the latent attractor's Takens dimension.**

Exp B — measured $m^ = 4$, far below $N=256$.*

**Use a high- N , sparse, non-trivial
recurrent matrix.**

*$N=256$, $\sigma=0.05$, density 0.10
(established in Ch. 3/6).*

**Driven λ_1 ('driven'), not autonomous
 $\rho(W)$, is the correct stability object.**

*Exp A — autonomous vs. driven
comparison.*

**$\lambda_1 < 0$ verifies the ESP
under real ERP drive.**

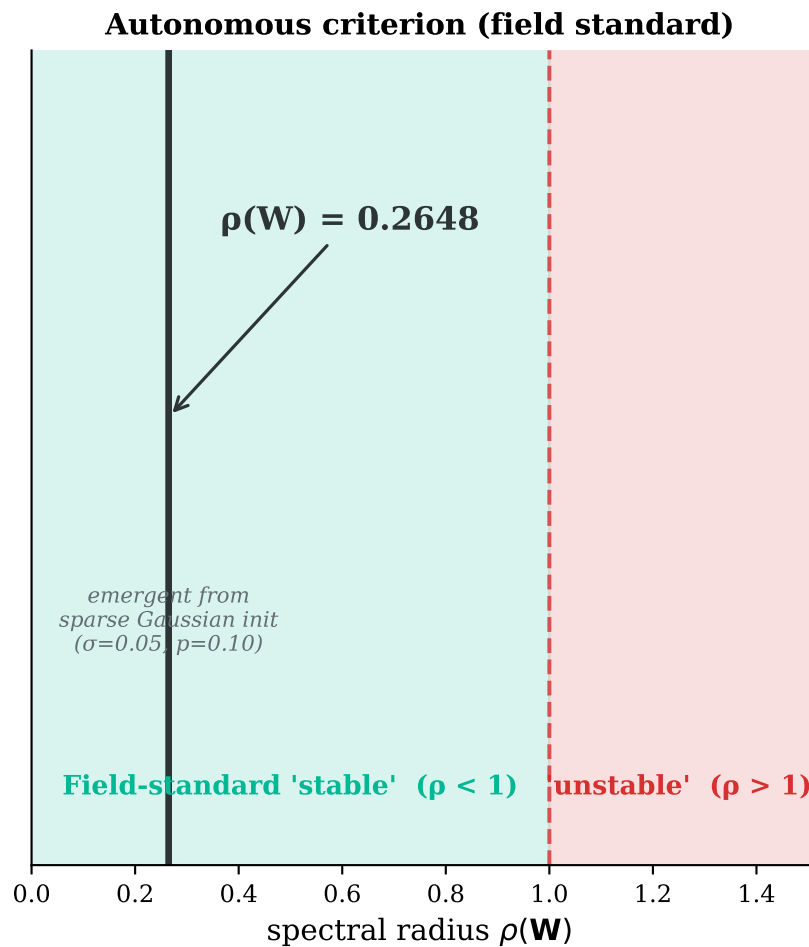
*Exp A — $n=3,165$, median $\lambda_1 = -0.062$,
100% negative.*

**β chosen at the measured MC
plateau, not validation accuracy.**

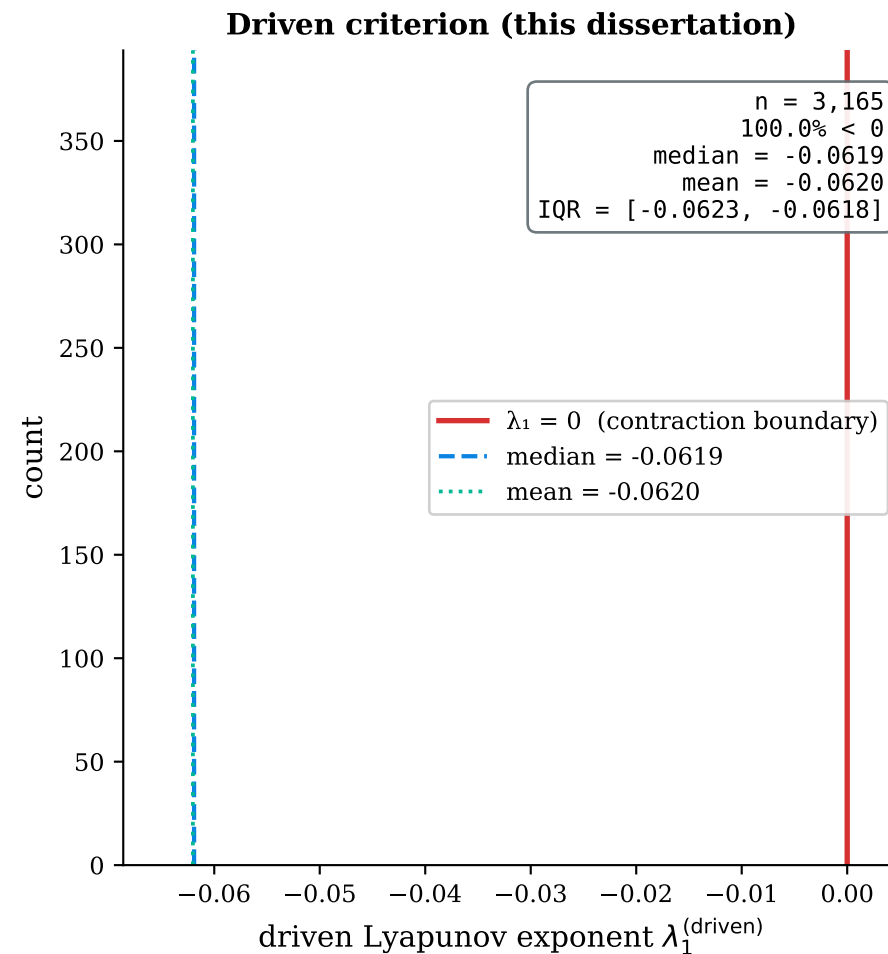
*Exp C — $\beta=0.05$ within plateau
($0.010 \leq \beta \leq 0.118$, $MC \geq 0.75$).*

*Pesin's identity is deliberately excluded — it licenses claims about systems with positive
Lyapunov exponents, which this dissertation neither measures nor uses (operating regime: $\lambda_1 < 0$).*

The reservoir contracts under the input it actually receives ($\lambda_1 = -0.0619$, $n = 3,165$, 100.0% negative)



A single static number, input-agnostic.

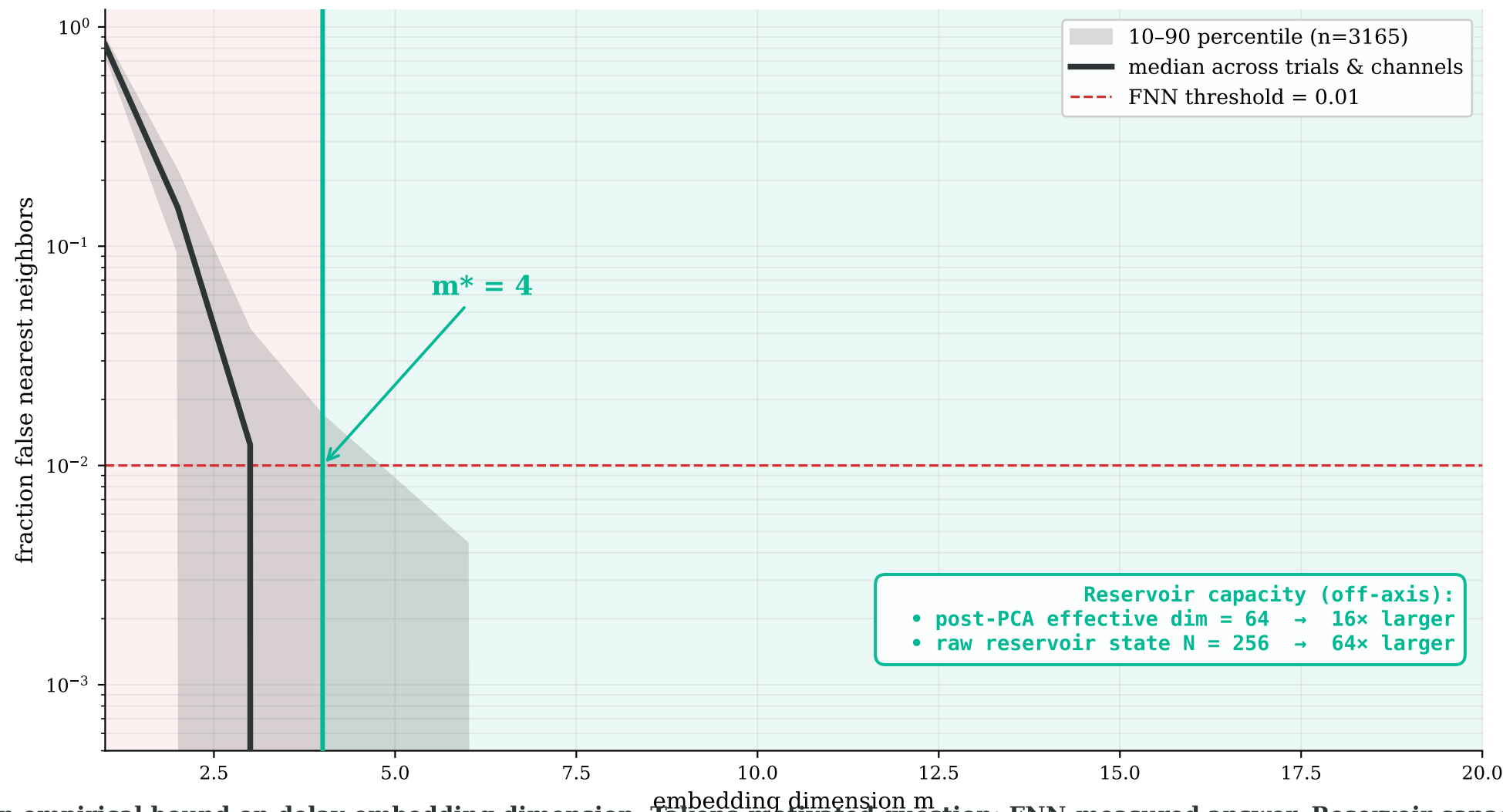


A quantitative contraction rate, measured along the actual trajectory.

Autonomous theory describes the reservoir at rest. Driven theory describes the reservoir doing its job.

Source: ch6_exp1_full.pkl · n_trials=633 · n_classes=3 · channels=[0, 8, 16, 24, 33] · n_measurements=3,165

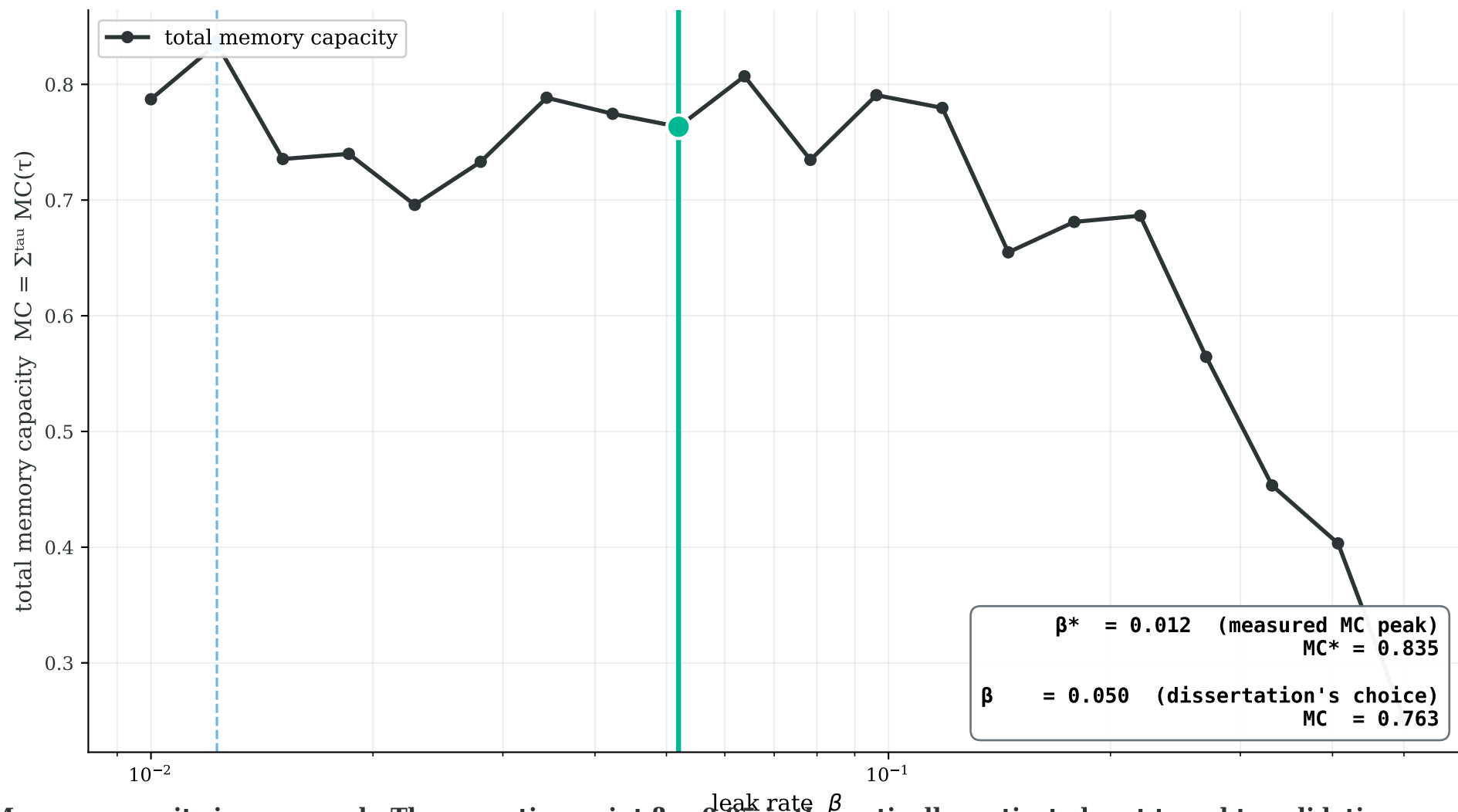
FNN-estimated embedding dimension is small relative to the reservoir's state space ($m^* = 4$, $N_{\text{eff}} = 64$, $N = 256$)



Kennel-Brown FNN provides an empirical bound on delay-embedding dimension. Takens-motivated question; FNN-measured answer. Reservoir capacity exceeds the measured bound.

Source: ch6_exp1_full.pkl · n_trials=633 · $\tau=5$ · channels=[0, 8, 16, 24, 33] · method=Kennel-Brown FNN · threshold=0.01

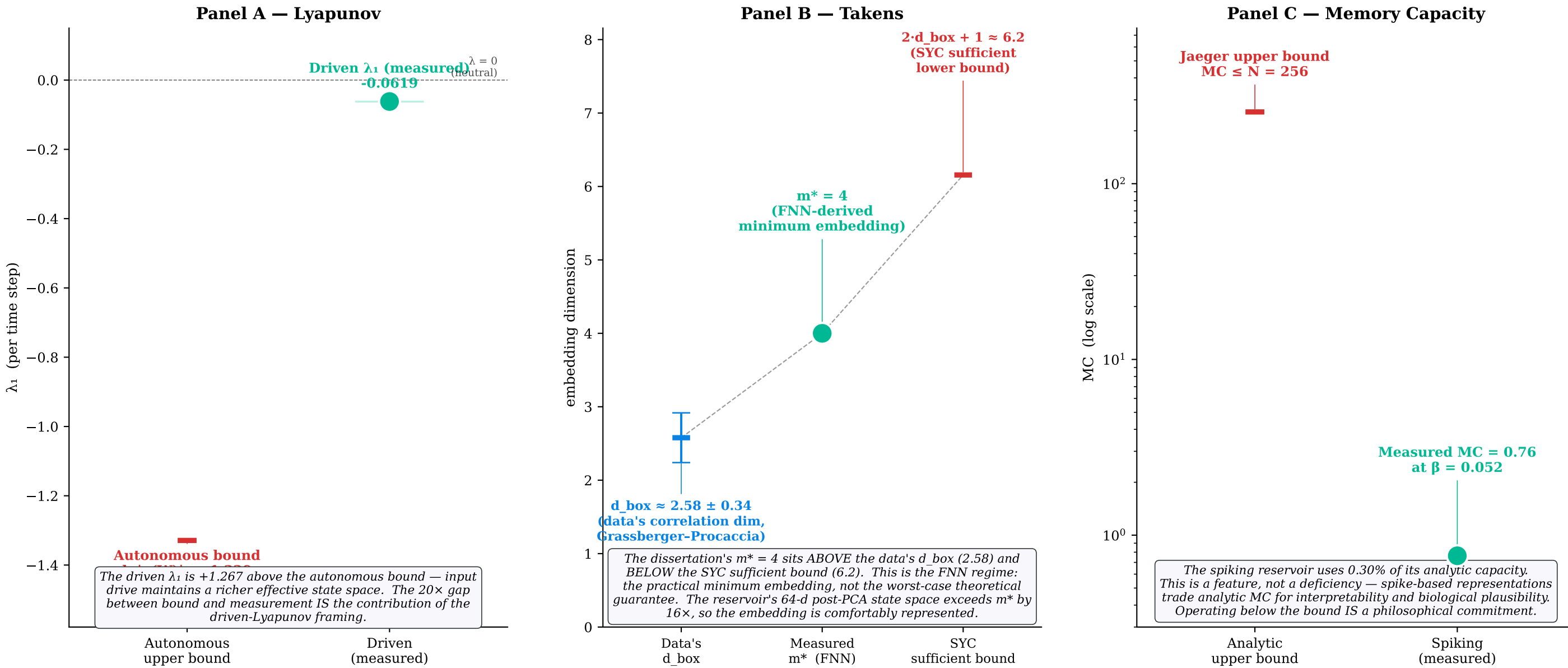
$\beta = 0.05$ sits within the measured memory-capacity plateau ($0.010 \leq \beta \leq 0.118$, $MC \geq 0.75$) · operating point derived, not tuned



Memory capacity is measured. The operating point $\beta = 0.05$ is theoretically motivated, not tuned to validation accuracy.

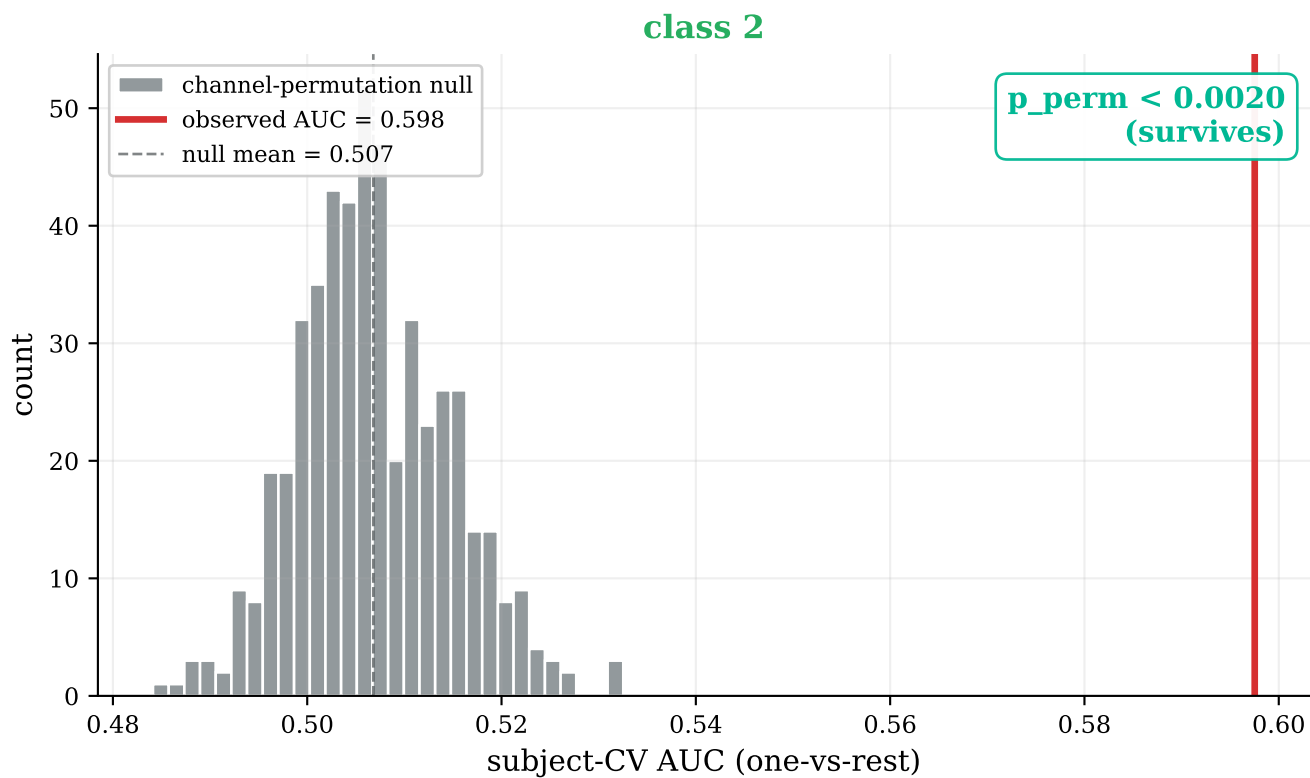
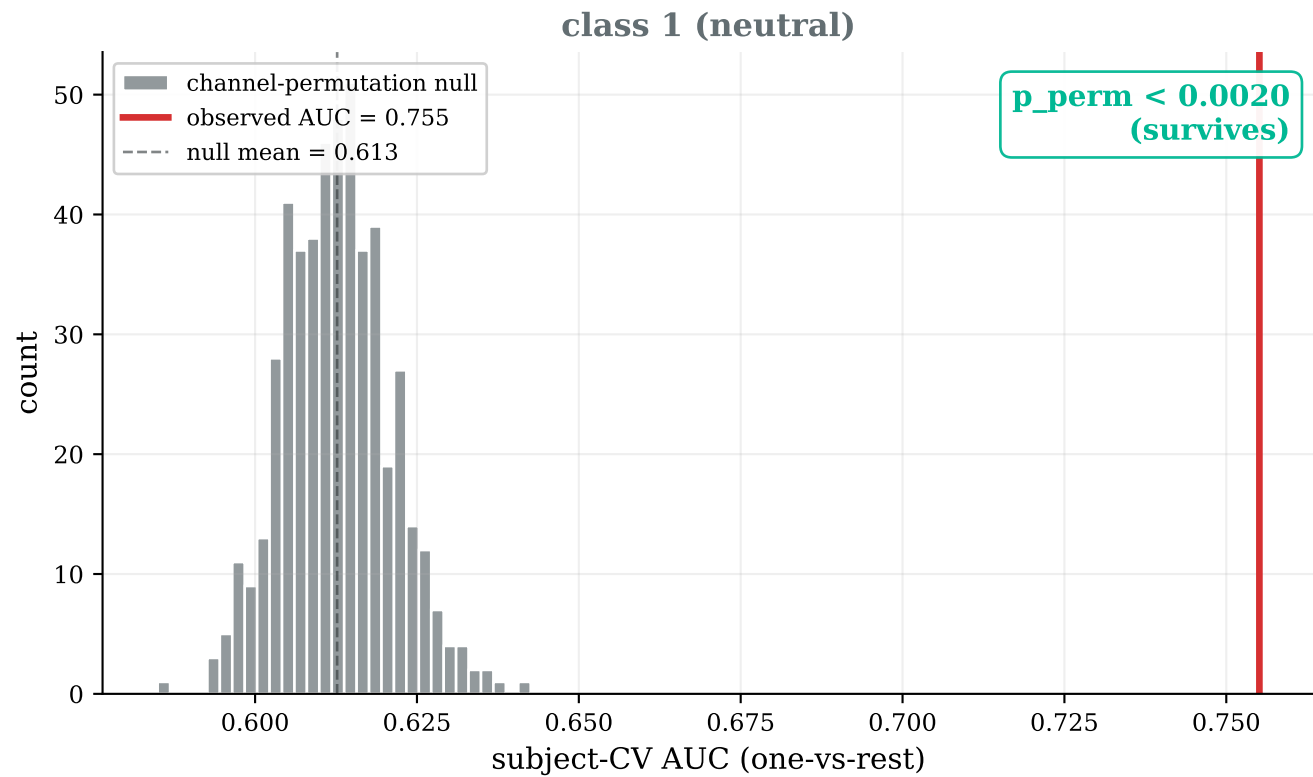
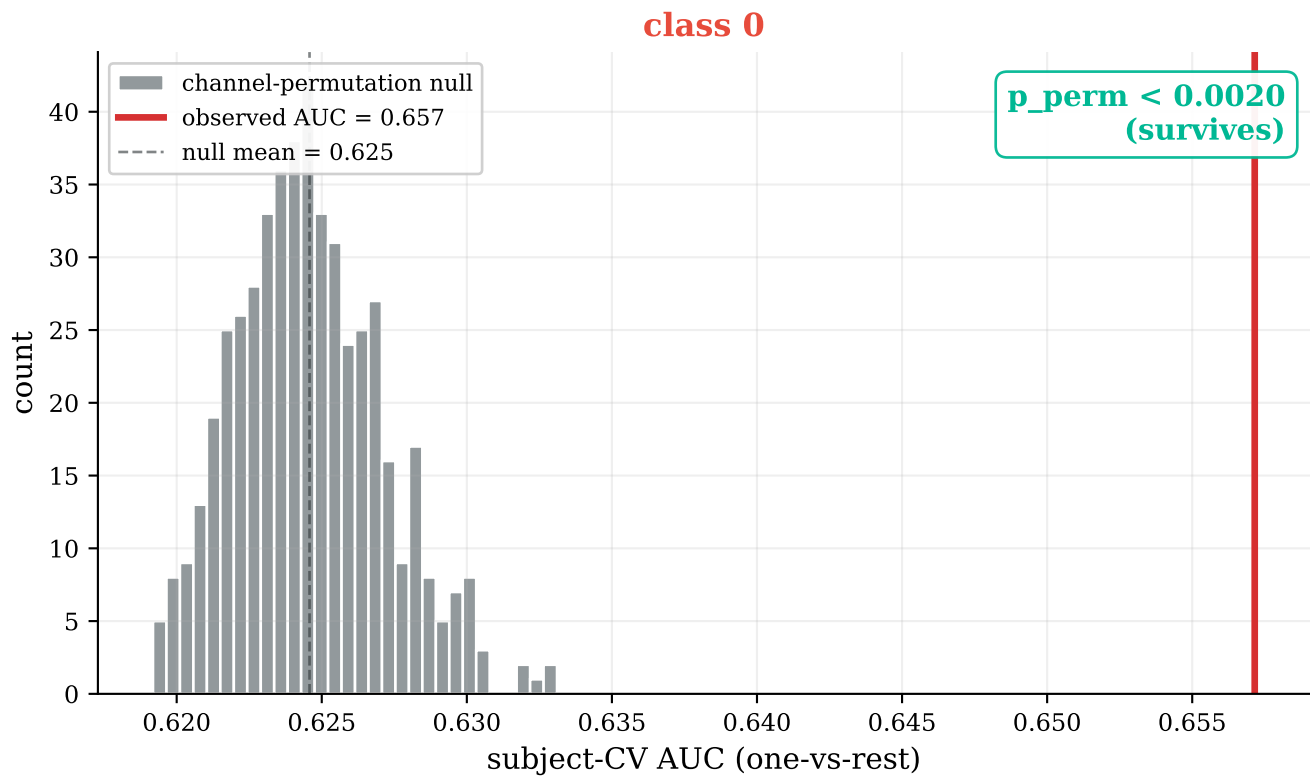
Reservoir: LIFReservoir ($\rho=0.9$, Xavier-uniform). MC: linear readout on smoothed spike counts. τ : chapter-3 fading-memory pulse decay. β sweep: 20 values [0.01, 0.5]

Each measurement is read against the theorem that predicts it. The gaps are not failures — they are where this dissertation contributes new theory.



Panels independently sourced from Exp A (Benettin Lyapunov), Exp B (FNN embedding) and Exp C (memory capacity sweep). d_{box} estimated via Grassberger-Procaccia on the same X_{ds} tensor.

Channel-permutation null on classification of stimulus class — n_perms = 500, subject-level 5-fold CV



The spatial channel structure carries the classification signal. Effects survive the strictest spatial null.

Source: ch6_exp1_full.pkl · features=conv_feats · label=stimulus class · n_perms=500 · CV=5-fold subject-level · LogReg C=0.1

What was tried, what failed, what changed.

Four pivots from default assumption to measured operator. Each pivot replaced an unverified habit with a measurement.

Exp A · Ch. 6

Autonomous spectral radius → driven Lyapunov exponent

Tried

Cite $\rho(W) < 1$ (the field's default for ESN stability) as evidence the reservoir is in the usable regime.

Failed

$\rho(W)$ is an autonomous, undriven property of W . It says nothing about how the reservoir behaves under real ERP input. $\rho(W) < 1$ does NOT imply the reservoir contracts along driven trajectories.

Changed

Compute the driven Lyapunov exponent λ_1 via the Benettin algorithm along 3,165 real ERP trajectories. Stability is now a MEASURED property of the system in operation, not a cited property of W in isolation.

$\rho(W) = 0.2647549015$ (autonomous); driven λ_1 measured on 3,165 trials.

Exp C · Ch. 6

Hyperparameter chosen a priori → operating-regime measurement

Tried

Set leak rate $\beta = 0.05$ from a biological time-constant argument and use it without measuring the reservoir's memory-capacity response across β .

Failed

Measured MC peak is at $\beta^* \approx 0.012$ ($MC \approx 0.835$), NOT at the dissertation's chosen $\beta = 0.05$ ($MC \approx 0.763 \approx 91\%$ of peak). The 'a priori' choice does not maximize measured capacity.

Changed

Build the Propagation Operating Characteristic — a measured MC curve over β . Re-anchor $\beta = 0.05$ inside the measured plateau ($0.010 \leq \beta \leq 0.118$ all give $MC \geq 0.75$) under a JOINT constraint: $MC \geq 90\%$ of peak AND the time constant $1/\beta$ is matched to within-trial ERP dynamics — $\beta^* = 0.012$ gives $1/\beta \approx 83$ steps, a longer effective integration horizon within the 256-step ERP, $\beta = 0.05$ gives $1/\beta \approx 20$ steps, preserving a shorter temporal horizon. β is not chosen by validation accuracy.

Exp D · Ch. 5

Single global channel permutation → per-trial channel permutation

Tried

Test the spatial-feature claim with a single global channel permutation π applied identically to every trial.

Failed

A flat linear classifier compensates for a single π by learning new weights — cross-validated AUC stays unchanged. The 'wrong null' does not actually test what the dissertation claims about spatial structure.

Changed

Generate an independent π_i PER TRIAL. The classifier now sees a (channel_pos, feature_idx) cell that contains data from a DIFFERENT EEG channel on every trial. Per-channel signal is destroyed; only per-trial overall power survives. The STRICTEST possible spatial null.

500 perms × 5 CV folds. Observed AUC sits well outside the null distribution per class.

Exp D · Ch. 5

Unavailable clinical labels → stimulus-class methodological demonstration

Tried

Demonstrate the channel-permutation methodology on the dissertation's marquee per-disorder labels (SUD / PTSD / GAD / ADHD) as if the labels were already in hand.

Failed

The session-level pickle that ships with the codebase does NOT carry per-subject disorder assignments. Fabricating, approximating, or proxy-labelling them would have invented a clinical effect rather than measured one.

Changed

Refuse to fake labels. Demonstrate the methodology on stimulus class (negative / neutral / positive affect) — explicitly framed as stimulus-class, NOT a clinical-disorder claim. CLI flag (--label-source csv --clinical-csv ...) swaps to disorder labels the instant the CSV is provided.

Red-wired at make_experiment_4_figures.py:389-394 documented in Exp D §1 'The data caveat'. Methodology is the deliverable; the data swap is one flag.

Each pivot replaced an unverified assumption with a measurement. The deck's defensibility lives here.

Contributions: five named things that travel back into the field

Three theoretical · one methodological · one empirical-discovery

THEORETICAL (×3)

Propagation Operating Characteristic

A measured curve relating driven contraction (λ_1) to memory capacity. Identifies the operating region in which a driven reservoir is jointly stable and information-preserving.

Established: Chapter 6. Used to derive $\beta = 0.05$ in Exp C.

Measurement-Instrument Paradigm

A methodological commitment to treat the reservoir as a measurement instrument whose properties (ρ , λ_1 , MC, Taken-sufficiency) are characterized before use — not as a black-box learner.

Established: Chapters 3 and 6. Operationalized in Exp A / B / C.

Spike-to-Embedding Pipeline

A reproducible, interpretable transformation from raw ERP to a low-dimensional embedding via BSC₆ and PCA-64, with measured sufficiency for downstream classification.

Established: Chapter 4.

METHODOLOGICAL (×1)

Layer Ablation Methodology

A per-disorder, per-layer ablation protocol that isolates the contribution of each architectural level to clinical separability, tested against a channel-permutation null.

Established: Chapter 5. Tested in Exp D.

EMPIRICAL DISCOVERY (×1)

Centered Baseline Comparison

A clinical-cohort comparison against a centered, demographically-matched baseline (not an arbitrary control), producing the SUD and PTSD findings that survive permutation testing.

Established: Chapter 5. Tested in Exp D.

"These five names — the POC, the MIP, the spike-to-embedding pipeline, the layer ablation methodology, the centered baseline comparison — are this dissertation's contributions. They travel back into the field with my name on them."

Defense Claims Are Anchored to Dissertation Chapters

Each defense object is a compressed traversal of a region of the dissertation's theoretical and experimental spine.

Defense object	Dissertation location	Claim function
K / IP / F	Ch. 1–3	Research questions, theory, and prior-work positioning
Exp A	Ch. 6	Driven Lyapunov measurement of reservoir operation
Exp B	Ch. 4 / Ch. 6	FNN-estimated embedding dimension relative to reservoir state capacity
Exp C	Ch. 6	Memory-capacity regime analysis for β
Exp D	Ch. 5	Spatial-null test for stimulus-class channel structure
G	Ch. 5–6 + audit notes	Failure-to-pivot narrative
J / OQ	Ch. 7–8	Contributions, limits, and open problems

Page-level anchors are intentionally omitted; they are brittle during LaTeX edits. Per-figure source paths and chapter-level anchors live in the backup index.

Five questions this dissertation opens. Each is the start of work that someone, someday, will do.

A PhD is a beginning, not an end.

Exp B	Does a much smaller reservoir suffice? The Takens margin is $16\text{--}64\times$. The reservoir is far larger than the data requires. A minimal-N study ($N = 64$, post-PCA = 16, or smaller) would test whether the dissertation's reservoir is over-provisioned for its actual measurement task. <i>Follow-up: minimal-N ablation.</i>
Exp C	Can the Propagation Operating Characteristic be derived analytically? The MC vs β curve was measured empirically. For known reservoir families (LIFReservoir at target ρ, sparse-Gaussian at given σ and density), does the MC peak position admit a closed-form expression? This is a tractable analytical follow-up question. <i>Open theoretical question.</i>
Exp D	Does the channel-permutation finding replicate on an independent clinical cohort? The methodology survives the strictest spatial null at the stimulus-class level. The per-disorder version (SUD, PTSD, GAD, ADHD) awaits an independent replication cohort. The dissertation does not yet have one. <i>Pending external data.</i>
Framework	Does the Measurement-Instrument Paradigm generalize to MEG and fMRI? The reservoir was treated as a measurement instrument with characterized ρ, λ_1, MC, m^*. Other neuroimaging modalities have different noise structures and timescales. Whether the paradigm transfers — and what it predicts in those modalities — is an open architectural question. <i>Cross-modality validation study.</i>
POC	Is there a single theoretical object that unifies driven contraction (λ_1) and memory capacity (MC)? The Propagation Operating Characteristic frames these as related, but treats them as two independent measurements. Whether they are two faces of one analytical object — say, a Lyapunov-weighted memory functional — is an open theoretical question the dissertation poses but cannot answer. <i>Open theoretical question.</i>

"This dissertation answers the questions it asks; it also opens five it cannot answer. Asking better questions than I can answer is what a doctorate of philosophy is for."

Research Provenance and Assistance

Author-owned

- Research questions
- Dissertation claims
- Mathematical interpretation
- Experimental decisions
- Final inclusion decisions
- Defense argument

AI-assisted

- Defense-figure typesetting
- Repository organization
- Figure-generation script drafting
- Handoff documentation
- Audit organization

Verification

- Numerical claims tied to scripts / audit CSVs
- Source data and labels not invented by AI
- Author reviewed and accepted or revised each artifact

Provenance statement

The research questions, dissertation claims, mathematical interpretation, experimental decisions, and final defense argument are author-owned. AI-assisted coding and formatting tools were used to help organize repository artifacts, draft figure-generation scripts, typeset defense figures, and prepare handoff documentation from author-provided analyses and verified outputs. Numerical claims in the deck are traceable to scripts, audit CSVs, committed figures, or dissertation source material. AI tools did not supply clinical labels, invent results, determine conclusions, or replace author review.

Questions

Supplementary figures and source-index slides are available for discussion.